Backend

Deployed · sensor intake verified

The Express API connects the board and owner dashboard to Supabase Postgres and Auth. It owns incident, approval, work-order, and verification state.

Deployment and API contracts

Vercel serves the React build and forwards /api routes to an Express handler. The handler uses Supabase Auth to validate owner sessions and the private bc schema to store state. Owner requests are checked against building membership. Device requests use a dedicated bearer credential and a fixed device identity, unoq-main.

Endpoint	Caller and behavior
GET /api/config	Public configuration for owner sign-in and app mode.
POST /api/hardware/telemetry	Authenticated UNO Q snapshot v1 for both buildings.
GET /api/state	Owner view with buildings, last readings, freshness, incidents, activity, and command records.
GET /api/buildings/:id/readings	Owner's latest ten minutes of readings, up to 300 samples.
GET /api/activity	Owner's recorded incident and service events.
GET /api/incidents/:id	Owner's incident details and current service state.
POST /api/service-requests/:id/decision	Owner supplies version and approved or declined.
POST /api/work-orders/:id/progress	Owner records contact, acceptance, findings, or verification.
/api/hardware/commands	Software command queue, device polling, and per-command acknowledgment. The current uploader does not consume it.

Telemetry, detection and storage

BACKEND

Record group	Current behavior
Device and building state	One device for office and data_center, separate membership records and sensor payloads.
Telemetry history	Deduplicate by device, boot ID and sequence. Reject captures over two seconds ahead or outside the 24-hour ingest window. Older and out-of-order samples do not overwrite current readings. Prune old telemetry periodically during ingest.
Freshness	A sample five seconds old or older is stale. A fresh sample after a gap records connection restoration.
Office low light	Requires expected and reported lighting on, a configured threshold, and three seconds below the threshold. Production threshold is unset. Local demo sets the ratio to 0.48.
Data-center flame/IR	Active or brief-activation flags from a fresh snapshot can open an incident. Keep one unresolved incident per building and category.
Temperature	Store raw thermistor ADC values and heat flags. The state and history APIs attach temperature estimates using per-building calibration. Temperature does not trigger incident rules.

The deployed reference is 71.7 °F with baselines of 235 and 395 counts. Raw counts are preserved. The room estimate assumes the KY-028 analog signal falls as temperature rises. A multi-point profile can replace the rough estimate when reference data is available.

Evidence to preserve

Incidents retain their original sensor evidence and source. Keep verified findings separate from inferred causes. Preserve accepted and verified repair records for future retrieval.

Approval gate and service lifecycle

BACKEND

01 A new incident receives a rule-based inspection proposal. The owner can approve or decline the current version before expiry. The decision checks membership, the version, and the incident state.

02 Approval commits the decision, one work order, one queued technician-contact action, and timeline entries in a transaction. An identical retry returns the existing decision. A conflict, expired request, or unauthorized request is rejected. A decline leaves the incident open.

03 The approved work order advances through queued, contacted, accepted, repair reported, and verified. The production owner records contact and acceptance manually. The local demo simulates those actions.

04 A report moves the incident to awaiting verification. The office can use three seconds of fresh in-range lighting or a manual check. The data center requires a recorded manual check. Verified work produces a repair case and resolves the incident.

Verification and next backend work

The automated tests cover membership, timestamps and freshness, deduplication, concurrent approval, decline, repair verification, and the software command round trip. The production database contains a hardware upload with sensor data for both models.

Keep the database connection, device credential, and future provider keys server-side. Give the browser only public Auth configuration. The local demo uses PGlite and simulated snapshots against the same schema and workflow.

Configure the office threshold after bench measurement. For Nemotron and Photon, add durable job and provider event records, verified inbound handlers, and a worker with retry and reconciliation. Keep the existing approval transaction as the authority for dispatch.

Repository and setup

Supabase project dashboard

Hosted API configuration